

SANTOSH SURESH MADANNAVAR

Belagavi, Karnataka | ✉ santoshmadannavar@gmail.com | 9880854939 | 🌐 Santosh02411 |
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Professional Summary

Results-driven **Software Engineer and Machine Learning Developer** with hands-on experience designing, building, and deploying end-to-end ML pipelines, RESTful APIs, and full-stack web applications. Proficient in **Python, Flask, React, XGBoost, and scikit-learn**, with applied expertise in **Generative AI, LangChain, and Retrieval-Augmented Generation (RAG)**. Delivered production-grade AI solutions through two professional internships including a 4-month AI Fellowship under VTU's Bharat Unnati program. Recognized top fellow for technical delivery and professional excellence. Seeking a software or data science role where I can build intelligent, scalable systems that create measurable business impact.

Work Experience

Data Science Intern

February 2026 – May 2026

EchoBrains Technologies Pvt. Ltd.

Bengaluru, Karnataka

- **Engineered 3 end-to-end ML pipelines** covering data ingestion, preprocessing, feature engineering, model training, evaluation, and Flask API deployment — reducing manual deployment effort by streamlining model-to-production workflows.
- Trained and optimized **supervised ML models** (regression and classification) using scikit-learn; achieved a **15% improvement in predictive accuracy** through systematic hyperparameter tuning and cross-validation.
- Built **interactive data visualization dashboards** using Matplotlib and Seaborn, surfacing actionable business insights from raw datasets and accelerating stakeholder decision-making.
- Deployed trained ML models as **production-ready REST APIs** using Flask, enabling seamless frontend integration and real-time inference for business-facing applications.

Artificial Intelligence Fellow

2025 – 2026

LearnersByte / ExpertPedia — Bharat Unnati TechLeap Program (VTU)

Hyderabad, Telangana

- Completed a **competitive 4-month AI fellowship** under VTU's Bharat Unnati initiative — one of select fellows chosen for applied industry training in Generative AI, Agentic AI, and advanced prompt engineering.
- Built **intelligent, context-aware applications** using LangChain, Retrieval-Augmented Generation (RAG), and large language model (LLM) APIs, including multi-step agentic workflows.
- **Recognized among the top fellows** for exceptional technical contributions, delivery quality, and professional conduct across all program milestones.

Education

Maratha Mandal's Engineering College

Belagavi, Karnataka

Bachelor of Engineering in Computer Science and Engineering

December 2022 – May 2026

- **CGPA: 8.67 / 10.0 (Overall)**
- **Relevant Coursework:** Python, C, Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks, Software Engineering, Machine Learning

Projects

Cold Weather EV Range Degradation Modeler | *XGBoost, Flask, SHAP, scikit-learn, Python*

2025

- **Developed an end-to-end ML web application** predicting EV battery range degradation under cold weather conditions, achieving **90%+ model accuracy** using XGBoost with extensive feature engineering on climate and telemetry datasets.
- Integrated **SHAP (SHapley Additive exPlanations)** for Explainable AI — delivering transparent, interpretable predictions with feature importance visualizations, boosting user trust in model outputs.
- Built a **Flask REST API backend** with a fully responsive HTML/CSS/JS frontend for real-time user interaction and on-demand inference.
- **GitHub** ↗

Unified Social Media Ads Platform | *Python, Flask, OAuth 2.0, REST APIs, SQLite*

2025

- Designed and built a **multi-platform advertising dashboard** supporting ad creation, publishing, and scheduling across **6 platforms** (Facebook, Instagram, LinkedIn, TikTok, Twitter, Pinterest) from a single interface.
- Implemented **OAuth 2.0 authentication** and integrated 6 third-party REST APIs for cross-platform campaign management and performance tracking, reducing manual ad management overhead.
- Architected a **modular Flask backend** with SQLite persistence and responsive frontend — designed for extensibility to add new platforms with minimal code changes.
- [GitHub](#) ↗

Malware Detection Using Machine Learning | *Python, XGBoost, Random Forest, SHAP, Flask*

2024

- Built a **cybersecurity ML pipeline** to detect and classify malware types using XGBoost and Random Forest trained on real-world security datasets — achieving high-precision threat identification.
- Applied **SHAP-based Explainable AI** to provide actionable threat-cause analysis, enabling end users to understand, audit, and act on detected security risks.
- Deployed as a **Flask web application** with an intuitive file-upload interface for real-time malware scanning and automated reporting.
- [GitHub](#) ↗

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++, HTML, CSS

AI / Machine Learning: scikit-learn, XGBoost, TensorFlow, OpenCV, SHAP, LangChain, Retrieval-Augmented Generation (RAG), Generative AI, Agentic AI, Prompt Engineering, Google Gemini API

Data Analytics & Visualization: Microsoft Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn

Web Frameworks & Libraries: Flask, Django, React, Node.js, Streamlit

Databases: MySQL, SQLite

Cloud & DevTools: IBM Cloud, Google Cloud Platform (Vertex AI), Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Postman

Core Competencies: REST API Design, OAuth 2.0, ML Pipelines, Feature Engineering, Explainable AI, Full-Stack Development, Data Structures & Algorithms, OOP, Software Engineering

Certifications & Training

The Ultimate Job Ready Data Science Course | *CodeWithHarry*

Feb 2026

- Comprehensive data science curriculum: Python, statistics, ML algorithms, and end-to-end project development.

Getting Started with Artificial Intelligence | *IBM SkillsBuild* | Credly: credly.com/badges/f075f0b1

Sep 2025

- IBM-issued badge covering foundational AI concepts, ML applications, and real-world AI use cases.

Retrieval Augmented Generation with LangChain | *Adobe Learning Manager*

Sep 2025

- Hands-on implementation of RAG pipelines using LangChain for context-aware, LLM-powered enterprise applications.

Machine Learning SkillUp | *Simplilearn*

Apr 2025

- Supervised and unsupervised learning, evaluation metrics, and practical end-to-end model building.

Journey to Cloud: Envisioning Your Solution | *IBM SkillsBuild* | Credly: credly.com/badges/f4e1a634

Sep 2025

- Cloud architecture fundamentals, solution design, and IBM Cloud platform essentials.

GE Aerospace Explore Engineering Job Simulation | *Forage*

Apr 2025

- Practical engineering simulation: energy source design, performance analysis, and compression ratio evaluation.

Languages

English (Professional Proficiency) **Kannada** (Native) **Hindi** (Proficient)